

Scaling

A brief guide to scaling forest businesses and organizations from local to landscape

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Abstract

Surprisingly, little is written on scaling nonprofits. Overwhelmingly, a lot is written on scaling businesses. By any measure it is not easy to do. We offer a background on scaling along with considerations, planning and budget in particular, to consider when scaling operations from the local to landscape. Broad success factors include clear leadership and planning, measuring standardized outcomes, and piloting successes may not translate or replicate across in other regions or landscapes. Recommendations from scaling at project outset, to realistically calculating the cost to scale are offered.

Keywords f, u, n, d, i, g, ., ., o, r, e, s, t, b, a, p, l

1. TAKEAWAYS

1. **Tailoring strategies.** Off-the-shelf solutions do not exist, especially in the forest health arena. Tailored strategies for scaling are therefore required, but bespoke solutions make scaling challenging, particularly if they have to be adjusted for each situation, geography, or organizational component.
2. **Pilots may not work.** Sometimes referred to as the “pilot and pray” approach. In many cases, we may be running out of time, funding, and resources to conduct pilots. However, if you have to pilot a project, try it at a small scale, with clear hypotheses and metrics set up to test effectiveness.
3. **Measuring outcomes.** Create monitoring and measurement systems that measure outcomes or impacts of forest health projects and your company’s success. See the [Monitoring](#) chapter for more details on creating a system to measure your business success.
4. **Workforce.** Finding qualified employees is a major challenge for the private and public sectors. Registered professional foresters (RPFs) are in short supply, and knowledgeable mill operators are difficult to find and train. Even assembling qualified thinning or prescribed fire crews led by experienced individuals can be difficult. Challenges are increased for nonprofits that offer lower salaries than the private sector.
5. **Leadership.** From an ownership perspective, company leadership can be tricky, e.g., letting go as the company founder when the business expands. Depending on organizational needs, leadership may be needed in different locations or over time.

2. BACKGROUND

Surprisingly, little is written on scaling nonprofits. Overwhelmingly, a lot is written on scaling businesses; much of it is breathless commentary about tech companies moving from a single founder to 1,000s of employees and how great the fireman's pole at the SOMA office in San Francisco is. Few to none of this *oeuvre* is linked to forest health, with most articles focused on scaling tech startups and most exploring scaling nuts and bolts.

2.a. Emotions

Molly Graham, from her experience of scaling at Google, Facebook, and Quip, takes a unique scaling approach, examining the people and emotional side of scaling, noting that you should not try avoiding or ignoring the emotions that couple with the scaling tornado, but acknowledging they're normal and any other organization will be going through similar circumstances Graham (2024). She says that emotional acknowledgment, which she likens to making friends with the monster chewing on your leg, is only half the battle; the other half is how you respond (Figure 1).



Figure 1: Acknowledging and responding to the monster chewing on your leg is critical for managing scaling. Graphic credit: Graham (2024).

Considering emotions in scaling is crucial for managing the workforce and also an important part of scaling leadership. Let's look at other approaches from the conservation and agricultural fields.

2.b. *Up/out/deep*

In the conservation arena, Salafsky et al. (2021) adapted a framework created by Moore et al. (2015) for scaling conservation nonprofits. Based on their literature review, there are three scaling approaches:

1. **Scaling out** involves replicating the initial pilot with three options: 1. expanding the scope, 2. replicating pilots within a program, and 3. promoting innovation diffusion by capturing and communicating what you've learned and getting other organizations to adopt your strategies.
2. **Scaling up**, e.g., policy change, takes a systems-level approach to leverage scale across a greater area rather than testing and replicating pilots.
3. **Scaling deep** is a behavioral change approach that changes the underlying values of the actors in a system. For example, it might involve building a stewardship ethic among landowners or creating and communicating stories that convey the needed values to key audiences. It's best to take a careful, measured approach to scaling deep. In many cases, this type of strategy takes a lot of time and requires many touch points and well-thought-out strategies to succeed. Simply running radio spots to bolster messaging isn't enough to change behavior.

Scaling out is common in conservation but has several downsides, including high risk and cost. Piloting may ultimately lead to what many conservation projects follow the 'pilot and pray' approach Salafsky & Margoluis (2021). In other words, implementing conservation actions based on effectiveness while ignoring scalability risks Pienkowski et al. (2024). A bottom line reality is that we may frequently be beyond the pilot stage for forest or biodiversity conservation due to the advanced state of wildfires, climate change, and habitat loss. Pienkowski et al. (2024) says the inverse can be true, e.g., promoting scalability over effectiveness, and that effectiveness can change with scale, possibly requiring different methods to measure outcomes. For example, voluntary carbon markets have failed to deliver benefits due to inadequate methods to measure additionality West et al. (2023).

2.c. *Farmer-2-farmer*

Compared to agriculture, conservation falls far behind in understanding the factors related to adoption and scale. In his seminal book on farmer-to-farmer extension called *Two Ears of Corn*, Roland Bunch describes scaling agricultural extension, starting small and spreads by diffusion Bunch (1982). One of the main components of this approach begins with small-scale experimentation. Farmers drive experimentation at this level on tiny, ~50 m², plots, creating a low-risk approach. Small-scale experimentation has several advantages at the programmatic level since it can reach the poorest farmers due to its low-risk, low-cost approach.

As more farmers do more experiments and successful strategies are chosen, the diffusion of the program spreads rapidly as farmers adopt cost-effective farming practices. The approach also has the advantage of being self-sustaining and does not need technical assistance from outside experts since farmers learn directly from one another; in other words, the program creates and turns its flywheel Collins (2005). Like the scaling deep approach mentioned above, farmer-to-farmer approaches can be scaled as movements and connected to social movements for food sovereignty Brescia (2017).

Brescia (2017) offers a similar scaling framework to Salafsky et al. (2021):

1. **Depth.** Groups of farmers innovate on their farms.
2. **Breadth.** Known as horizontal scaling and can be achieved when practices are spread across many households and communities.
3. **Verticality.** Comes into play when enabling farming practices across networks and movements and linking community farmers to markets and supportive policies.

2.d. *Planned & scaled funding*

Funding is an important factor in scaling not covered in the agricultural examples above. In the check your scaling assumptions Salafsky et al. (2021) provides an excellent example of developing realistic budgets based on strategic plan goals and costing those out over time. They show that to reach a seemingly realistic but ambitious goal of 500,000 tons of CO₂e sequestered, a hypothetical organization will have to carry out 250 projects over a decade. This is when s*%t gets real! They then create a financial model to determine total projects, costs, tons of carbon sequestered, and cost/ton. The upfront work demonstrates the value of testing assumptions made to carry out a plan Salafsky et al. (2021). A similar analysis could be completed simultaneously, focusing on staffing and workforce needed to carry out the plan.

3. FOREST BUSINESSES

Scaling businesses is a double-edged sword: while it is a well-trodden path with numerous success stories, the pressure to scale begins immediately, accompanied by expectations for profit and system-wide success. Factors business owners may want to consider when planning to scale include

- **Finance.** As a business scales, Different timings, amounts, and for different items may occur, further complicated by changes in markets and other external factors. Financing from traditional institutions for new products and approaches may take longer to develop but should become easier as more forest health businesses come online with successful track records.
- **Funding.** A business mantra for funding could be to use government grants to test and loans to scale. Government grants can be used to test new ideas, practices, and equipment and act as a powerful innovation and test phase before scaling. Phased properly, along with purposeful testing, grants can allow pilots to be rapidly scaled. Following successful test implementation, loans can scale

landscape implementation following the proven track record developed with agency grants. As highlighted in [Program Design](#), use your business or strategic plan to prioritize the grants you pursue rather than allowing grant programs to change your priorities and vision.

- **Leadership/ownership.** Company leadership can be tricky from an ownership/ideas perspective, e.g., letting go as the business expands. New leadership may be needed to manage satellite businesses and hire the right leadership to maintain corporate culture. Company founders may be used to overseeing all aspects of project and company development, so delegating those tasks and creating the structure to do it effectively becomes more critical as a company grows. Succession as a company matures carries the same needs and cannot be overlooked, particularly during rapid growth. Similarly, over-dependence on the vision of the original leader/founder can hamper a company's ability to scale and be successful beyond the original business bounds.
- **Markets.** Many forest product markets are new, making them difficult to create. Creating a cooperative across multiple businesses to weather product startups may help, or entering new markets slowly, deliberately, and at small scales until they are established or you have consistent buyers is another. Another way may be to diversify your product lines with a mix of traditional items to create a core revenue source and new products to test their selling power and establishment.
- **Quality control.** Maintaining quality control with franchises can be difficult, especially when the original product artisans can no longer oversee every aspect of production. Training new staff for satellite production facilities should be done using the scaling procedure and given enough time. Quality control can also be compromised when rapid production increases are required. Keeping pace with this change is difficult to predict but must be factored into expansion or scaling plans.
- **Maintenance.** If you're focused on forest health and restoration, maintaining previously treated sites should be built into your scale model with workforce, transportation, equipment use, and distance to sites for processing wood all considered. Maintaining equipment, vehicles, and other hardware in your business depends on what needs to be built into your costs and how you scale and have the right people to maintain needed equipment schedules.
- **Sustainability.** How sustainable is your business as it expands? Are there controls and principles in place to maintain sustainability? Creating and maintaining a focus on a triple bottom line, measuring your company's performance based on profit, people, and planet (or environmental factors), building in B-corp certification, or giving back to the community can be significant factors. At another level, how do all of the factors in this section help your business weather change over time, and how do you measure sustainability or resilience to changes in supply, production, and markets?
- **Measuring outcomes.** Create monitoring and measurement systems that measure the outcomes or impacts of forest health projects and your company's success.

4. CHALLENGES

Rapid scaling is a huge challenge, particularly for nonprofit organizations that are often unable to scale rapidly due to funding, workforce, and governance issues. These issues seem overwhelming, but each piece is not insurmountable and may frequently have simple solutions. Patterns of challenges for scaling include

- **Workforce.** Finding qualified employees is a major challenge for the private and public sectors. RPFs are in short supply, and knowledgeable mill operators are difficult to find and train. Putting together qualified thinning or prescribed fire crews led by experienced individuals is challenging. Challenges are increased for nonprofits that offer lower salaries than the private sector.
- **Chicken and egg finance.** Banks are no longer experienced in-house with the due diligence required to review loans for forest health and infrastructure projects. To compensate, they ask businesses to return once the business is functioning. This chicken and egg situation, akin to not being able to apply for a job out of college w/o experience, is trying for new businesses or existing businesses trialing new technologies. This challenge may decline over time but certainly does not help accelerate the pace and scale of forest health treatments at the present moment.
- **Funding & scale.** Funders often ask for scalable projects but don't provide the necessary funding to scale once a project is successful. Also, public investment through grants, which is fantastic for trialing new and innovative programs or equipment, is not being leveraged with private funding.
- **Nonprofits scaling pace.** Nonprofits typically do not scale quickly. It's impossible to scale when you fund many projects based on a reimbursable format, such as funding from most state and federal agencies.
- **Agency agreements impede scale.** Currently, the slow speed of executing agency agreements—some may take upwards of one year to follow grant awards—significantly impedes the pace and scale of forest health implementation. The cost and speed of environmental compliance are similar issues, putting the brakes on project implementation.
- **NIMBY.** Neighbors may oppose projects despite the need for biomass utilization, forest products, and mills. For example, Alpenglou Timber in Truckee, California, in late 2024, had a NIMBY challenge even though some of the project opponents were on biomass promotion projects or boards!
- **Project size $\neq$ need.** Coordinating across multiple actors in a region can be particularly challenging and take a lot of time, time that some organizations do not have when starting new entities. For example, projects in the California Wildfire Task Force tracking dashboard may look small compared to the actual goal of treating 1 million acres throughout the state.
- **One size doesn't fit all.** This makes for a conundrum that prevents scaling since every effort to scale requires bespoke changes in policy, funding, workforce, behavioral, and institutional change Churches (2023). Furthermore, some companies may be content to scale to a certain size; others may wish to grow further. This right size can be factored into planning by the organization's

board, but bottoms-up ownership of this growth also needs attention, not just top-down direction from the board.

- **Effectiveness & scalability.** Evidence of scaling effectiveness, readiness, and risk evaluation can be lacking in projects. The actual total impact is not dependent on scalability alone, it is the sum of effectiveness and scale Pienkowski et al. (2024).

5. RECOMMENDATIONS

Overall, strategies need to be tailored. Considerations to incorporate scale when developing your business, project, and programs for forest health:

- **Scale from the outset.** Consider scaling from the start of your program design.
- **Calculate scale costs.** Rough out the costs of scaling your work and assess feasibility.
- **Emotional scale.** Be aware of the emotional cost of scale to you as a business owner or nonprofit manager and its impact on your staff.
- **Replication & adoption.** Replication and adoption are moving targets and change as you scale.
- **Effectiveness.** Measuring success becomes more complicated with scale but infinitely more valuable for learning and determining what is working or not.

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